AGRIAI INTELLIGENCE SYSTEMS | CROP PROTECTION PLATFORM

Cloud & Container Deployment Guide

Docker, Docker Compose, Streamlit Cloud, Render, and Railway

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Executive Summary: This document provides official specifications, standard operating procedures, and architectural blueprints for the AI-Powered Crop Pest Prediction & Agricultural Advisory System. The platform bridges machine learning multi-class diagnostics, agro-meteorological streaming data, and agronomic Integrated Pest Management (IPM) to protect farm yields sustainably.

1. Containerized Deployment (Docker & Compose)

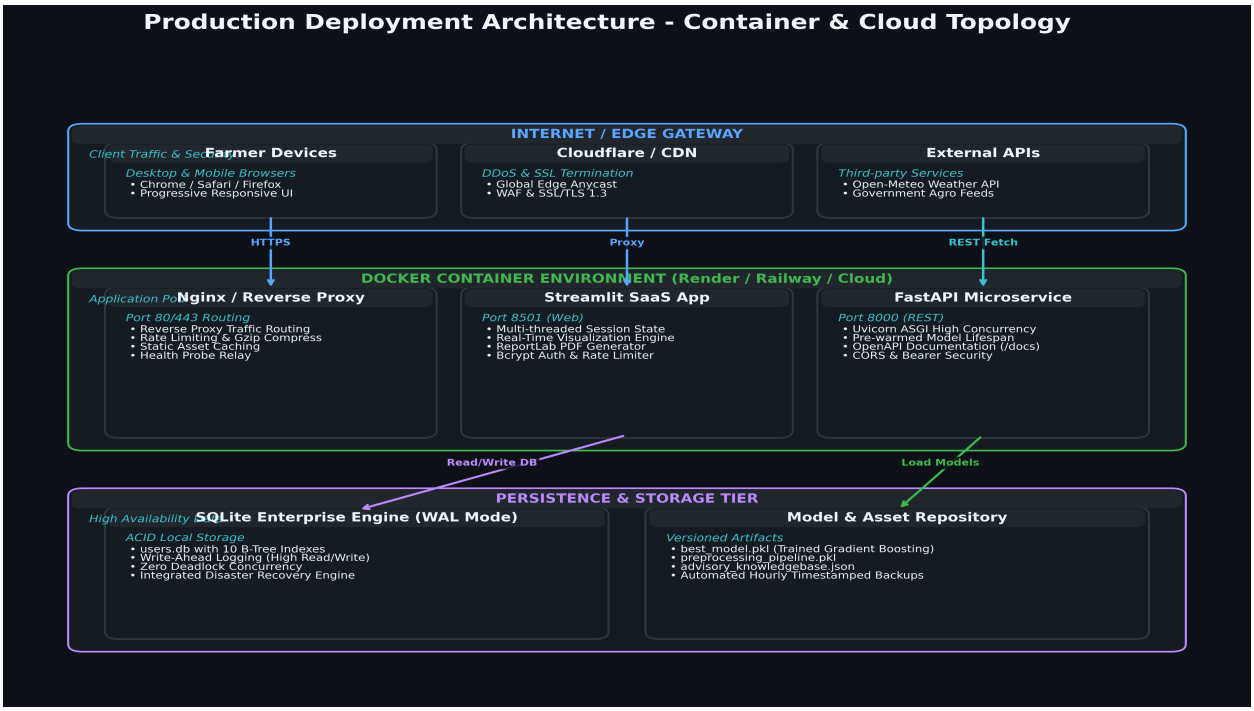
The application includes an enterprise-ready, multi-stage Dockerfile adhering to least-privilege security principles.

Building & Launching with Docker Compose:

```
# Build image and launch web container in background:
docker-compose up -d --build

# View live logs:
docker-compose logs -f web

# Verify container health:
docker-compose ps
```



2. Cloud Deployment Targets

Platform	Type	Build Command	Start Command
Streamlit Cloud	Serverless PaaS	Automatic via requirements.txt	app/app.py
Render	Web Service	pip install -r requirements.txt	streamlit run app/app.py --server.port $PORT
Railway	Container / Procfile	Auto-detected Dockerfile	Procfile entry: web
AWS EC2 / VPS	Linux Virtual Machine	pip install -r requirements.txt	systemd service + Nginx reverse proxy

3. Production Health Checks & Monitoring

Automated heartbeat probes are available on both interfaces:

- Streamlit Health Probe: http://localhost:8501/_stcore/health
- FastAPI Health Endpoint: http://localhost:8000/health